

Environmental Inversion: A Collection of Artworks that Subvert Methods of Climate Change Action

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ABSTRACT

This paper details Environmental Inversion, a collection of artworks that propose a critical examination of climate change, waste creation and removal. “Inversion” is defined and its relevance to climate change research, followed by detailed descriptions of projects by the author that examine this exploratory integration. Each project consists of an energy source that reverses its intended use. Examples include creating pollution through green energy to contaminating air through filtering hardware powered by fossil fuels. The works act as a critical reminder of attempts to greenify our planet, why they are pursued, and the energy-wasteful methods employed to reach these solutions.

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1 INTRODUCTION

Pollution and the use of unsustainable energy sources have become major threats to our way of living. In particular, the problem of excessive garbage or waste contamination is global and will take the effort of every person on the planet to help rectify. Proclaiming how prevalent this problem is within the United States, Pulitzer Prize winning author Edward Humes in his book, “Garbology: Our Dirty Love Affair With Trash,” describes the plethora of trash generated by each individual and how this amount impacts our continued legacy on earth. “Americans make more trash than anyone else on the planet, throwing away about 7.1 pounds per person per day, 365 days a year. Across a lifetime that rate means, on average, we are each on track to generate 102 tons of trash. Each of our bodies may occupy only one cemetery plot when we’re done with this world, but a single person’s 102-ton trash legacy will require the equivalent of 1,100 graves. Much of that refuse will outlast any grave marker, pharaoh’s pyramid or modern skyscraper: One of the few relics of our civilization guaranteed to be recognizable twenty thousand years from now is the potato chip bag.” [Humes, 2012] One enduring problem with environmental waste production and

removal is that the processes used to eliminate and recycle trash generates enough detritus to harm the environment even further. Whether this is exhaust from the trucks that gather garbage and recyclable materials to the contaminants that escape the air from the incinerators [Baptista, 2019] meant to make trash “disappear,” these methods of “clean up” only exacerbate the problem. Environmental Inversion is a collection of works by the author that aim to challenge how we as a society are reacting to waste removal and clean up as well how our methods of generating supposedly clean energy often backfires to create the opposite effect.

2 BACKGROUND

Inversion, defined as a “reversal of position, order, form, or relationship” [Webster’s Dictionary, 2023] refers to changing how one thinks about an object in order to invoke the opposite of that relationship. Directing something to behave in a way that is contradictory from its intended use is a pivotal ingredient to effective design thinking and innovation. Also known as “TRIZ” or the “Theory of Inventive Problem Solving” [Altshuller, 1999], coined in 1946 by Russian scientist Genrich Altshuller, one of its key principles is that there are fundamental contradictions in the solution to most problems and that by eliminating those contradictions in design, solutions become easier to grasp. Countering this process, Environmental Inversion proposes the position that by introducing a reversal of traditional roles and integrating them back into design principles, novel approaches to solve these issues could emerge. In the early 1800s, German mathematician Carl Jacobi followed the principle of “man muss immer umkeren” [Clear, 2015] translated as “invert, always invert” which was a strategy of invoking clarity in thinking by solving math problems in reverse form, thus solving the opposite of the problem first. To this day, inversion thinking remains a powerful methodology for creatively solving complex problems by thinking both forward and backwards for solutions. Similarly the concept of “reverse engineering” is often used to understand the complexities of an electronic device in order to understand how it functions by deconstructing it from the inside out. In their book, “Reversing: Secrets of Reverse Engineering”, [Eilam and Chikofsky, 2015] Chikofsky and Eilam describe “reverse engineering” as “the process of extracting the knowledge or design blueprints from anything man-made” and claim that the concept dates back to the beginning of the industrial revolution in the 1760s. They go on to explain that the difference between reverse engineering and scientific research is “that with reverse engineering the artifact being investigated is manmade, unlike scientific research where it is a natural phenomenon.” The projects in Environmental Inversion stand as a form of reverse engineering by using one energy source to unravel another in order to shed light on their overall use.

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Figure 1: *Tree Logic*, N. Jerimijenko, 1999. (©1999. Photo: K. Sharon)

3 RELATED WORK

The field of environmental media art has a rich history and broad spectrum of approaches and methodologies that span its storied existence. Examining artwork that specifically focuses on ecological disaster and climate change as it relates to computer hardware and software, there are several works that the author drew inspiration from in creating *Environmental Inversion*. Natalie Jerimijenko's two projects, *One Tree* and *Tree Logic* [Jerimijenko, 1999] both provoke novel methods for thinking about how exposure to environmental pollutants and direct physical circumstances effect living creatures, in her case, plants. *Tree Logic* relates to the theme of "inversion" consisting of several trees suspended upside down and forced to grow towards the earth, instead of their natural inclination to grow upwards towards the sun. This shift in position inverts a plant's basic instruction set, resulting in an example of reimagining the innate capabilities of natural creatures. Similarly, her *OneTree* [Jerimijenko, 2023] project examines how a cloned tree planted in different locations around the metropolitan San Francisco area responds differently to environmental stimuli and natural conditions within its immediate locations. This concept relates to the idea that by shifting relationships between natural objects and their current environments, we can understand more about how and why they function. Juxtaposing climate change volatility and using it to create sustainable action is Julian Oliver's *Harvest* [Oliver, 2017] that consists of a wind turbine that when spun, generates power to run a computer that is used to mine for cryptocurrency. After the crypto is created it is used to fund research efforts that aim to curb climate change. Thus a direct relationship is created between the climate itself and how we are helping or hurting it becomes evident in the work. Similarly, Philip Phelan's "CO2nvert" series of projects integrate ethics into traditionally "dumb" electronics such as washing machines and toasters forcing them to only function when the power they are using originates from "sustainable energy" sources. For instance, "Snobby Toaster" [Phelan, 2001] will only toast your breakfast if it detects that the power it consumes derives from green sources such as solar, wind, or hydro-electric. These small changes in design show how critical thinking is challenging the ways in which artists and designers are modifying the landscape of sustainable ecological practice.



Figure 2: *Snobby Toaster*, P. Phelan, 2002. (©2002. Photo: P. Phelan.)

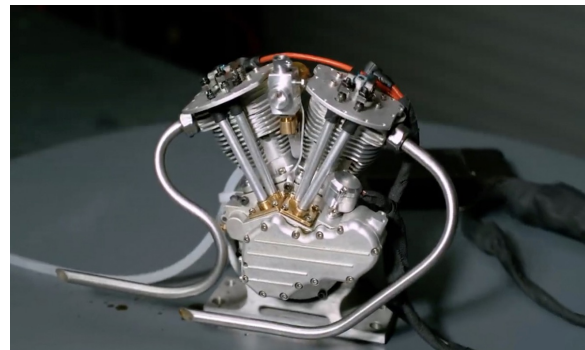


Figure 3: *Clean Smog Panhead Engine* (©2023)

4 PROJECTS IN THE ENVIRONMENTAL INVERSION SERIES

Projects included in the *Environmental Inversion* series examine themes of ecological waste and its subsequent removal. These include *Clean Smog*, a gas-powered engine that creates electricity to power an air purifier which then cleans the pollution it creates, and *Loss Depletion*, a solar powered CO2 dispenser that uses sustainably sourced power to re-pollute the atmosphere with CO2. *Take Out The Trash* is a downloadable MacOS application that when opened, simply fills up the users trash can at infinitum with garbage files. This action forces them to "empty" it which yields no consequence since the application simply keeps filling it continuously.

5 CLEAN SMOG

Over the course of a year, one vehicle on the road is capable of emitting up to 4.6 metric tons of carbon dioxide while every gallon of gasoline that that vehicle consumes creates about 8,887 grams of CO2. [Eilam and Chikofsky, 2015] This alarming amount of pollution is exacerbated by the fact that over 286 million carbon-emitting vehicles are on the road daily with only about one percent of vehicles being clean emissions vehicles in the United States. In a 2018 study [Jerimijenko, 2023] of 135 participants examining

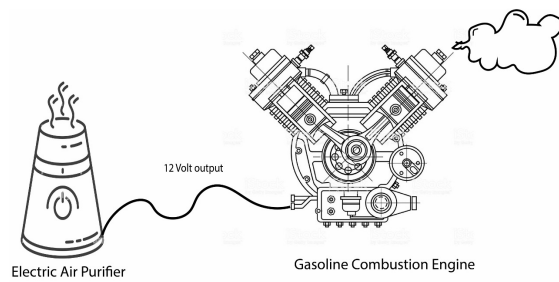


Figure 4: Clean Smog System (©2023)

cardiovascular responses to walking and engagement in healthy activities such as running or jogging, it was found that undertaking in these types of activities along heavily polluted streets actually cancelled out the benefits associated with these exercises. Building on this theme of direct environmental cancellation due to humanity's reliance on energy generation from fossil fuels, Clean Smog is a combustion powered air purifier artwork that simultaneously causes and removes its own pollution.

The project intends to rescind pollution by creating power through dirty / gasoline driven fuel that when burned, powers a generator that creates electricity to run an air purifier which ultimately extracts the same pollution from the surrounding air. The project is a comment on how anti-pollution methods often fail because the technology used to bring them to market causes more pollution than its intended goal. Clean Smog's hardware consists of a gas powered panhead combustion engine that when turned on, powers a DC power supply connected to a consumer air purifier. Clean Smog examines our relationship to the planet and climate change from a critical standpoint. This aspect of the work engages with civic, social, intellectual, communal, political, artistic, and environmental factors combined. How we got here, where are we going, who can we rely on to get there, and what has brought us to this point of reflection and questioning of our fragile futures as a global society? All of these questions reveal that we rely daily on technology that rarely improves our lives beyond what we already can maintain. Is the solution to a clean environment to use pollution emitting vehicles and dirty power generated electricity that ultimately contributes to more pollution?

6 LOSS DEPLETION

Also reflecting this theme is Loss Depletion, a solar powered carbon dioxide emitter artwork. Loss Depletion is a critical examination of the advent and mass adoption of seemingly "green" energy solutions meant to help our planet avoid the impending threat of global warming that could lead to mass extinction. Despite a push towards "clean" or green energy solutions, this shift can often lead to an increase in carbon emissions caused by the machinery and technology needed to make this transition into a reality. In 2022, the United States Energy Information Administration reported that 22% of all electrical power in the US was generated by clean renewable energy sources while 60% was still created by fossil fuels [Phelan, 2001]. Despite an urgent need to transition to sustainable energy

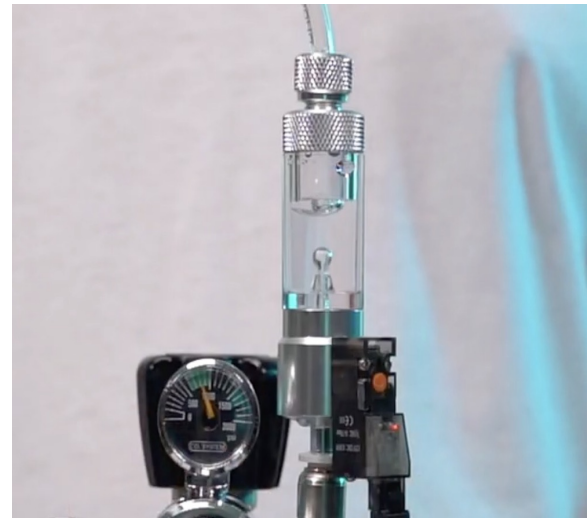


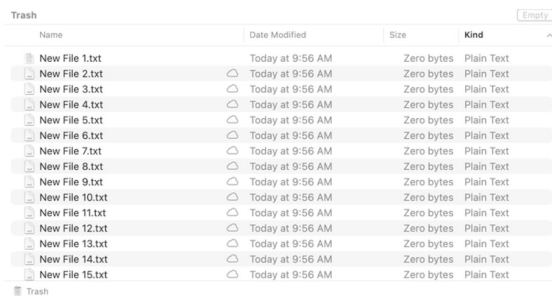
Figure 5: Loss Depletion (©2023)

practices, there is still a reluctance due to cost, availability, and government regulations. Loss Depletion provokes and challenges the dilemmas surrounding the global move to sustainable energy sources by using a natural power source, the Sun, to cause, not deter, global pollution. The device consists of an off the shelf ZRDR CO2 Generator System with a pressure gauge and automatic pressure relief valve with a bubble counter that is typically found in aquarium systems. Generating CO2 in the environment is typically a byproduct of moving heavy machinery and equipment since those combustion powered devices require fuel to function. Natural forms of CO2 emitters include the ocean, organic matter such as plants, trees, animals, and most of all, humans, who add over 30 gigatons of CO2 annually to the earth's atmosphere [Phelan, 2001]. This excess pollution is impossible for our land and oceans to absorb, causing it to trap heat in the atmosphere and ultimately lead to global warming and climate disaster. Loss Depletion comments, reacts, and acts as a pivotal indicator of humanity's inability to curb climate change despite its best intentions to do so.

7 TAKE OUT THE TRASH

Take out the Trash is a desktop MacOS application that creates a new text file and automatically moves it to the trash at infinitum.

As we use computers daily, we are often inundated with countless files on our computer that take up more space than necessary. Despite our intention to deal with these files, they are typically never deleted due to factors such as negligence, laziness, or a simple disregard for computer maintenance. Take Out the Trash, like its namesake, simply creates more trash and asks you to empty the trash in order to remove this unwanted waste. If one begins to empty the trash before the application is finished its cycle, it will still create trash even while it is being emptied. This process parallels society's "throw away" culture or the creation of objects such as electronics, appliances, and packaging that are meant to be discarded rather than be improved or recycled. A survey of 2,000 people from 2020 found that 52% of them have never deleted any



Name	Date Modified	Size	Kind
New File 1.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 2.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 3.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 4.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 5.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 6.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 7.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 8.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 9.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 10.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 11.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 12.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 13.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 14.txt	Today at 9:56 AM	Zero bytes	Plain Text
New File 15.txt	Today at 9:56 AM	Zero bytes	Plain Text

Figure 6: Take Out The Trash / Screenshot of Digital Trash (©2023)

files, apps, or anything else from any of their devices. Another 33% said they have files on their phones or other devices that are ten or more years old. If one parallels this act of hoarding in digital space to that in physical spaces [Jerimijenko, 1999], it was found that hoarding [Humes, 2012] is an activity found in 2 to 5 percent of all American adults. Houses and living spaces fill up with seemingly unwanted objects and possessions, most of which could be classified as throwaway junk. The act of digital hoarding goes even further in that with “digital hoarding, however, the act of saving the file becomes an uncontrollable urge” [Jerimijenko, 1999] so much that choice vanishes and an overpowering need surpasses want or an understanding of what is necessary. These types of digital versions of what is more commonly found in the physical world can be troubling for even the most advanced computer users and for those who regularly exhibit these activities without really understanding why they are occurring.

8 FUTURE PROJECTS IN THE SERIES

While examining ways in which human activity physically pollutes the atmosphere and planet, most sources derive from heavy machinery, refuse, and the domestication of livestock for work and food. Future projects in the Environmental Inversion series include HydroAir, a wind powered hydroelectric generator that uses one natural power source of wind to create power from water pressure or hydroelectric means. This type of misuse of “clean energy” comments on the overuse and wastefulness of some supposedly green energy solutions that often come with an unintended price. In addition, Re:Pollute is a critical gas-powered device equipped with a CO₂ smog sensor so that when smog is detected, the device creates more pollution to add to the greenhouse effect. The goal of Re:Pollute is to solely contribute to global warming despite the obvious inclination to do the opposite.

9 CONCLUSION

Artworks in the Environmental Inversion series examine the principle of inverting a given form of energy production in order to counteract its intended goal. Beginning with Jacobi’s problem solving principles of inversion where it is often more telling to begin at the end of a question than focus on its starting point, each work in the series maintains its focus of subverting current modes of environmentalist thinking by turning these solutions on their heads



Figure 7: Take Out The Trash Application Icon (©2023)

and forcing them to counter their original or intended goals. Each of the Environmental Inversion artworks are aimed to shed light on and act as critical instigators to our continued reliance on fossil fuels to satisfy humanity’s growing energy needs and consumption. Each project utilizes and questions an energy source which then is inverted to cause or create more pollution or waste removal.

For example, Take Out The Trash reminds us of how the phenomenon of hoarding [Humes, 2012] has infiltrated our digital lives to the point where seemingly endless cloud storage solutions have made us disregard the need to be frugal about what files or digital detritus are important to us or not. The idea that “wastefulness” is something that could be merely disregarded is a novel phenomenon in the computer industry as extended storage capabilities have become so cheap that we no longer have to think about where our files are being kept is something that was not even considered a few years ago. The overall goal with the projects is for them to act as starting points for discussions about how climate change is effecting global resources and how to invert the methodologies used to stem its rapid gains.

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